Solver Documentation

electroChem

PotentialReaction branch

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1 Introduction

This document covers the custom OpenFOAM solver electroChem made for simulating reacting multicomponent potential flows, with the goal of simulating microbial electrosynthesis of CO_2 to acetate. The solver is an extension of the existing pimpleFOAM FVM solver from OpenFOAM.

1.1 PotentialReaction branch

This branch restructures the code and couples equilibrium reactions and potential corrections to the transport equation ensuring the solution is converged on the transport path. The reaction setup is restructured to treat general equilibrium reactions and include them as source terms.

2 Governing equations

The flow is treated as incompressible and solved through the incompressible Navier Stokes formulation using the PIMPLE algorithm

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho}\nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g} \tag{2}$$

All active species concentrations are treated by reactive Nernst-Planck transport. Reaction source and potential are iteratively corrected using Picard iterations, iterating the solution over k steps while time is held fixed until he concentration and potential fields are mutually consistent with both electro-neutrality and current conservation. Only when the Picard loop is converged does the solver proceed in time n. Each Picard iteration starts using the potential and equilibrium source from the previous time-step $(\phi^{(1)}, \lambda^{(n)})$

$$\frac{\partial c_i^{(n+1)}}{\partial t} + \nabla \cdot \left(c_i^{(n+1)} \mathbf{u} \right) = \nabla \cdot \left(D_i \nabla c_i^{(n+1)} \right) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i^{(n+1)} \nabla \phi^{(k)} \right) + R_i^{(k)} \tag{3}$$

Where c is the concentration of species i, $\mathbf{u}$ is the velocity, D is the diffusion coefficient, z is the charge, F is the Faraday constant, R the universal gas constant, T the temperature, ϕ the potential and R_i the combined contribution of fast and slow reactions

$$R_i^{(k)} = R_{kin.i} + R_{eq.i}^{(k)} \tag{4}$$

Slow reactions are treated kinetically by

$$R_{kin.i} = \sum_j \nu_{ij} \dot{r}_j \tag{5}$$

where ν_{ij} is the stoichiometric constant of species i for kinetic reaction j. Fast equilibrium reactions are assumed to instantaneously settle into equilibrium within a time-step and $R_{eq.i}$ determine the required extend of the equilbirum reaction to satisfy local equilibrium

$$R_{eq.i}^{(k)} = \sum_r \nu_{ir} \lambda_r^{(k)} \tag{6}$$

Where ν_{ir} is the stoichiometric coefficient of species i for the equilibrium reaction r and λ is an equilibrium pseudo source. The first Picard iteration starts from the potential of the previous physical time-step, $\phi^{(1)} = \phi^{(n)}$, with no equilibrium correction, $\lambda_r^{(1)} = 0$. Subsequent Picard iterations use the corrected potential and equilibrium source from the preceding Picard iteration. The Picard loop is evaluated using the equilibrium residual

$$F_r = P_r^{(k)} - K_r R_r^{(k)} \tag{7}$$

where K_r is the equilibrium constant of reaction r and P_r and R_r are the product and reactant contributions to the equilibrium equation

$$P_r^{(k)} = \prod_{\nu_i > 0} c_i^{(n+1)^{\nu_i}}, \quad R_r^{(k)} = \prod_{\nu_i < 0} c_i^{(n+1)^{-\nu_i}} \tag{8}$$

Electro-neutrality is evaluated using the net charge density

$$E = \sum_i z_i c_i^{(n+1)} \tag{9}$$

Using these criteria normalized residuals are defined for equilibrium and electro-neutrality

$$\epsilon_{F.r} = \frac{|F_r|}{\max(|P_r^{(k)}|, |K_r R_r^{(k)}|, F_{abs.r})}, \quad \epsilon_E = \frac{|E|}{\max(\sum_i |z_i| c_i^{(n+1)}, E_{absTol})} \tag{10}$$

This provides an absolute and relative convergence criteria for both equilibrium source and electro-neutrality

$$(|F_r| \leq F_{abs.r} \quad \vee \quad \epsilon_{F.r} \leq F_{relTol}) \quad \wedge \quad (|E| \leq E_{absTol} \quad \vee \quad \epsilon_E \leq E_{relTol})$$

If both are satisfied the current equilibrium source $\lambda^{(k)}$ and potential $\phi^{(k)}$ are satisfactory and the solution can proceed in time. If any of the equilibrium criteria are violated the equilibrium source is corrected for the next Picard iteration $k+1$ with

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha_\lambda \Delta \boldsymbol{\lambda} \tag{11}$$

where the correction is found from

$$\mathbf{J}_\lambda^{(k)} \Delta \boldsymbol{\lambda} = -\mathbf{F}^{(k)} \tag{12}$$

see appendix A.1.2. If any of the electro-neutrality conditions are violated the potential is likewise corrected for the following Picard iteration $k+1$

$$\phi^{(k+1)} = \phi^{(k)} + \alpha_\phi \Delta \phi \tag{13}$$

Derived from the current-conservation equation

$$\nabla \cdot (\mathbf{J}_D^{(k)} - \kappa^{(k)} \nabla \phi) = 0 \tag{14}$$

with diffusive flux

$$\mathbf{j}_D^{(k)} = -F \sum_i z_i D_i \nabla c_i^{(n+1)} \tag{15}$$

and conductivity

$$\kappa^{(k)} = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{(n+1)} \tag{16}$$

From this the matrix system for potential correction $\Delta \boldsymbol{\Phi}$ is established

$$\mathbf{A}_\phi^{(k)} \Delta \boldsymbol{\Phi} = \mathbf{r}_\phi^{(k)} \tag{17}$$

see appendix A.2. When converged species are converted to mass fraction

$$Y_i = c_i \frac{M_i}{\rho} \tag{18}$$

and the inert species "I" is explicitly found by

$$Y_I = 1 - \sum_{i \neq I} Y_i \tag{19}$$

pH is the log concentration of H^+ in mol/L

$$pH = -log_{10}\left(\frac{c_{H^+}}{1000}\right) \tag{20}$$

3 Custom boundary conditions

This section covers the custom boundary condition and helper functions added with this model.

3.1 Molar concentration

The following boundary conditions describe direct Dirichlet molar concentration boundaries.

3.1.1 Charge balanced concentration

This boundary condition determines the concentration of a selected charged species such that the prescribed inlet composition satisfies the charge balance

$$\sum_i z_i c_i = 0 \tag{21}$$

For the selected balancing species b, the required boundary concentration is

$$c_b = -\frac{1}{z_b} \sum_{i \neq b} z_i c_i \tag{22}$$

where z_i is the charge number of species i. The concentrations of the other species define the charge imbalance that the balancing species compensates. The selected species must have $z_b \neq 0$, and the prescribed composition must permit a nonnegative value of c_b. This condition can only apply to one species pr patch.

3.1.2 Inert molar concentration

This custom boundary condition is used for the inert species and sets the inlet concentration as the remainder such that the total mass-fraction of the system adds up to 1. The remainder for the inert species "I" is defined

$$Y_I = 1 - \sum_{i \neq I} Y_i \tag{23}$$

Concentration fields are converted to mass-fraction using

$$Y_i = \frac{c_i M_i}{\rho} \tag{24}$$

followed by

$$c_I = \frac{\rho Y_I}{M_I} \tag{25}$$

So directly

$$c_I = \frac{\rho}{M_I} \left(1 - \sum_{i \neq I} \frac{c_i M_i}{\rho} \right) \tag{26}$$

It is noted that this condition can only apply to one species pr. patch. If Inert molar concentration and charge balanced concentration are applied to the same patch, the inert species must be charge neutral $z_I = 0$.

3.1.3 pH molar concentration

This custom boundary automatically determines the distribution of H^+ and OH^- ions at a given pH. c_{H^+} is directly found from pH using concentration in mol/L

$$c_{H^+} = 10^3 \cdot 10^{-pH} \tag{27}$$

and c_{OH^-} by

$$c_{OH^-} = \frac{K_W}{c_{H^+}} \tag{28}$$

The user points the BC to the relevant equilibrium model from which K_W is read to ensure internal consistency.

3.2 Molar Flux

This section covers boundaries for molar concentration flux

3.2.1 Galvanostatic molar flux

The concentration gradient at the boundary is formulated as a robin boundary condition consisting of a faradaic diffusion and migration contribution

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{app}}{D_i |\nu_e| F} - \frac{z_i F}{RT} c_i \nabla \phi \cdot \mathbf{n} \tag{29}$$

where $\mathbf{n}$ is the unit normal vector, with positive values meaning flux out of the control-volume, ν is the stoichiometric coefficient with positive value meaning production of species i or electrons e and j_{app} is the magnitude of the galvanostaticly applied current, see appendix A.3.1. Special cases for this boundary condition is for non-charged species reaction with the electrode, charged species that does not interact with the electrode and species who are neither charged or interact with the electrodes

$$\nabla c_i \cdot \mathbf{n} = \begin{cases} \frac{\nu_i j_{app}}{D_i |\nu_e| F} & \text{when } z_i = 0 \\ -\frac{z_i F}{RT} c_i \nabla \phi \cdot \mathbf{n} & \text{when } \nu_i = 0 \\ 0 & \text{when } z_i = \nu_i = 0 \end{cases} \tag{30}$$

In the first case only the faradaic and diffusion contributions affect the species, for the second only the migration and for the last a simple zero gradient condition is sufficient.

3.3 Potential Flux

The last section covers potential flux boundary conditions

3.3.1 Galvanostatic potential flux

The potential gradient boundary is derived from the current expression (see appendix A.3.1) to a Neumann boundary condition

$$\nabla \phi \cdot \mathbf{n} = \frac{\mathbf{J}_D \cdot \mathbf{n} - j_{n.app}}{\kappa} \tag{31}$$

which determines the potential that balances the diffusive flux and conductivity to the applied current. $j_{n.app}$ is the direction aware current density, see appendix A.3.1.

3.3.2 Zero current potential flux

This boundary condition sets the normal ionic current density to zero by balancing the diffusive and migration current contributions. From the current expression

$$\mathbf{j} = \mathbf{J}_D - \kappa \nabla \phi \tag{32}$$

the condition $\mathbf{j} \cdot \mathbf{n} = 0$ gives the Neumann boundary condition

$$\nabla \phi \cdot \mathbf{n} = \frac{\mathbf{J}_d \cdot \mathbf{n}}{\kappa} \tag{33}$$

where $\mathbf{J}_D$ is the diffusive current density, κ is the ionic conductivity and $\mathbf{n}$ is the outward unit normal. The diffusive current and conductivity are supplied by the potential model, preserving the lagged coupling within each Picard iteration.
A zero potential gradient is sufficient only when the normal diffusive current is also zero. This condition instead allows a potential gradient that produces an equal and opposite migration current. When the potential model is disabled, it reduces to a zero-gradient condition.

4 Reaction Models

The solver combines all reaction contributions into a single source-term for each species. This combines the contributions of slow kinetic reactions and fast equilibrium reactions (equation 4)

$$R_i = R_{kin.i} + R_{eq.i}$$

4.1 Kinetic Reactions

Kinetic reactions typically develop on the order of or much slower than that of the transport. Therefore they are captured kinetically by their reaction rates (equation 5)

$$R_{kin.i} = \sum_j \nu_{ij} \dot{r}_j$$

Currently the model supports reversible kinetic reactions using known forward and backwards reaction rates and multiplicative Monod used microbial consumption and production.

4.1.1 Reversible kinetic reaction

The reaction rate of reversible kinetic reactions are found as

$$\dot{r} = k_f \prod_{\nu_i<0} c_i^{-\nu_i} - k_b \prod_{\nu_i>0} c_i^{\nu_i} \tag{34}$$

where k_f is the forward reaction rate and k_b is the reverse reaction rate.

4.1.2 Multiplicative Monod reaction

Microbial reactions are described through multiplicative Monod

$$\dot{r} = X \dot{q}_{max} \left(\frac{c_d}{c_d + K_{S.d}} \right) \left(\frac{c_a}{c_a + K_{S.a}} \right) \tag{35}$$

where X is the biomass density, $\dot{q}_{max}$ is the maximum specific consumption rate, K_S is the half saturation constant and the subscripts d and a describe the electron donor and acceptor substrates.

4.2 Equilibrium Reactions

Fast equilibrium reactions that are much faster than that of transport and slow kinetic reactions can be assumed to instantaneously settle into equilibrium within each solver time-step. This allows for treatment of fast reactions without resolving the kinetic rates which becomes very stiff for fast reactions. Instead they are incorporated as an equilibrium source term for each species telling how far the equilibrium reaction must proceed to satisfy the equilibrium condition of the system (equation 6)

$$R_{eq.i} = \sum_r N_{ir} \lambda_r$$

This is done for a general setup supporting a system of fast equilibrium reactions. For a given set of equilibrium reactions, all participating species are collected in the concentration vector of N_s species

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} \tag{36}$$

The combined stoichiometric coefficients $\mathbf{N}$ for all species N_s of all reactions N_r is defined

$$\mathbf{N} = \begin{bmatrix} \nu_{11} & \nu_{12} & \cdots & \nu_{1r} \\ \nu_{21} & \nu_{22} & \cdots & \nu_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ \nu_{N1} & \nu_{N2} & \cdots & \nu_{Nr} \end{bmatrix} \tag{37}$$

From the equilibrium reaction

$$K = \frac{\prod c_P^{\nu}}{\prod c_R^{\nu}} \tag{38}$$

The equilibrium residual equation system is formed

$$\mathbf{F}(\mathbf{c}) = \begin{bmatrix} \prod c_{p.1}^{\nu} - K_1 \prod c_{R.1}^{\nu} \\ \prod c_{p.2}^{\nu} - K_2 \prod c_{R.2}^{\nu} \\ \vdots \\ \prod c_{p.r}^{\nu} - K_r \prod c_{R.r}^{\nu} \end{bmatrix} = \mathbf{0} \tag{39}$$

From this the concentration Jacobian is defined

$$\mathbf{J}_c = \frac{\partial \mathbf{F}}{\partial \mathbf{c}} \tag{40}$$

As the equilibrium reactions are treated by a source term the concentration Jacobian cannot be used directly but must be translated to a source response. This is done using the spatial transport-response $\mathbf{H}_i$ (see appendix A.1.1) this becomes the equilibrium Jacobian

$$\mathbf{J}_\lambda = \mathbf{H}_i \sum_r \nu_{ir} \tag{41}$$

Then the spatially coupled equilibrium problem simplifies to

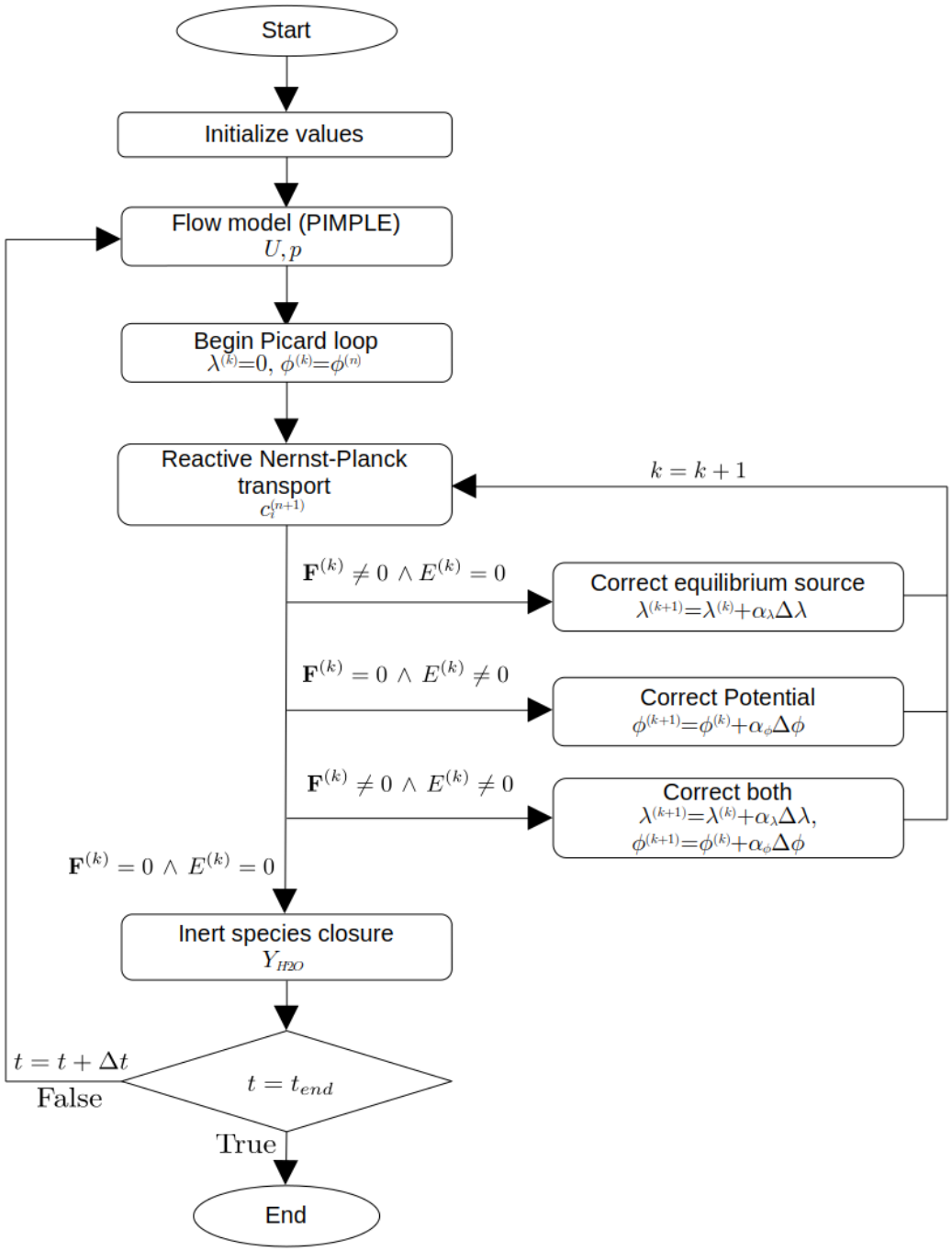
$$\mathbf{J}_\lambda \Delta \boldsymbol{\lambda} = -\mathbf{F} \tag{42}$$

See appendix A.1.2. The Newton correction $\Delta\boldsymbol{\lambda}$ determines the increment to the equilibrium reaction-rate fields $\boldsymbol{\lambda}$ that will satisfy the equilibrium condition

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha_\lambda \Delta \boldsymbol{\lambda} \tag{43}$$

where α_λ is a relaxation factor ($0 \leq \alpha_\lambda \leq 1$).

5 Algorithm flowchart



6 Solver structure

Source code structure

```
electroChem/
|
|-- Documentation/
|
|-- Make/
|   |-- files
|    options
|
|-- reactionModels/
|   |
|   |-- reactionManager/
|   |   |-- reactionManager.C
|   |   |-- reactionManager.H
|   |
|   |-- fastEquilibriumReaction/
|   |   |-- fastEquilibriumReaction.C
|   |   |-- fastEquilibriumReaction.H
|   |
|   |-- multiplicativeMonod/
|   |   |-- multiplicativeMonod.C
|   |   |-- multiplicativeMonod.H
|   |
|   |-- reversibleKineticReaction/
|       |-- reversibleKineticReaction.C
|       |-- reversibleKineticReaction.H
|
|-- potentialModels/
|   |
|   |-- potentialManager/
|   |   |-- potentialManager.C
|   |   |-- potentialManager.H
|   |
|   |-- electroneutralGalvanostatic/
|       |-- electroneutralGalvanostatic.C
|       |-- electroneutralGalvanostatic.H
|
|-- electroChem.C
|-- createFields.H
|-- UEqn.H
|-- pEqn.H
|-- cEqn.H
|-- YEqn.H
|-- diagnostics.H
```

custom functions

```
src/
|-- customLibs/
|   |-- electrochemistry/
|       |-- derivedFvPatchFields/
|           |-- MolarFlux/
|           |   |-- galvanostaticMolarFlux.C
|           |   |-- galvanostaticMolarFlux.H
|           |
|           |-- PotentialFlux/
|           |   |-- galvaniostaticPotentialFlux.C
|           |   |-- galvaniostaticPotentialFlux.H
|           |   |-- zeroCurrentPotentialFlux.C
|           |   |-- zeroCurrentPotentialFlux.H
|           |
|           |-- MolarConcentration/
|               |-- pHMolarConcentration.C
|               |-- pHMolarConcentration.H
|               |-- inertMolarConcentration.C
|               |-- inertMolarConcentration.H
|               |-- chargeBalancedConcentration.C
|               |-- chargeBalancedConcentration.H
|
|-- Make/
    |-- files
    |-- options
```

case file

```
case/
|-- 0/
|   |-- U
|   |-- P
|   |-- phiE
|   |-- c_*
|-- constant/
|   |-- electroChemistry
|   |-- reactions
|   |-- transportProperties
|   |-- turbulenceProperties
|-- system/
    |-- controlDict
    |-- fvSchemes
    |-- fvSolution
```

constant/reactions/reactions structure

```
reactions
{
    <reactionName>
    {
        type    <reactionModel>;

        stoichiometry
        {
            <species> <signed coefficient>;
            ...
        }

        // model-specific parameters
    }
}
```

A Appendix

A.1 Fast equilibrium reactions

A.1.1 Transport response

Starting from reactive transport

$$\frac{\partial c_i^{(n+1)}}{\partial t} + \nabla \cdot \left(c_i^{(n+1)} \mathbf{u} \right) = \nabla \cdot \left(D_i \nabla c_i^{(n+1)} \right) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i^{(n+1)} \nabla \phi^{(n)} \right) + R_{kin.i} + R_{eq.i} \tag{44}$$

This is rewritten as

$$\mathcal{T}_i(c_i, \phi) = R_{kin.i} + \sum_r N_{ir} \lambda_r \tag{45}$$

where $\mathcal{T}_i$ represents transient and Nernst–Planck transport, ν_{ir} is the stoichiometric coefficient of species i in equilibrium reaction r, and λ_r is the unknown volumetric reaction rate required to satisfy equilibrium through transport. After spatial discretization and linearization, the species equation over the domain becomes

$$\mathbf{A}_i \mathbf{c}_i = \mathbf{b}_i + \mathbf{V} \sum_r N_{ir} \boldsymbol{\lambda}_r \tag{46}$$

where $\mathbf{A}_i$ is the assembled transport matrix, $\mathbf{b}_i$ contains the known contributions, and $\mathbf{V} = \mathrm{diag}(V_1, \ldots, V_{N_c})$ contains the cell volumes. The vectors $\mathbf{c}_i$ and $\boldsymbol{\lambda}_r$ contain the concentration and equilibrium reaction rate in all N_c cells. In the implemented transport equation, transient, convection and diffusion contributions are assembled implicitly, while migration and kinetic reaction rates are evaluated explicitly from the available iterate. The transport matrix used for the response includes the applied matrix relaxation and constraints. To determine how the concentrations respond to an equilibrium-source correction, consider the perturbations

$$\boldsymbol{\lambda}_r \to \boldsymbol{\lambda}_r + \Delta \boldsymbol{\lambda}_r \tag{47}$$

The concentration will respond accordingly by

$$\mathbf{c}_i \to \mathbf{c}_i + \Delta \mathbf{c}_i \tag{48}$$

Holding the transport matrix and known contributions fixed during this correction gives

$$\mathbf{A}_i(\mathbf{c}_i + \Delta \mathbf{c}_i) = \mathbf{b}_i + \mathbf{V} \sum_r N_{ir}(\boldsymbol{\lambda}_r + \Delta \boldsymbol{\lambda}_r) \tag{49}$$

Subtracting the unperturbed equation yields

$$\mathbf{A}_i \Delta \mathbf{c}_i = \mathbf{V} \sum_r N_{ir} \Delta \boldsymbol{\lambda}_r \tag{50}$$

The concentration response is therefore

$$\Delta \mathbf{c}_i = \mathbf{A}_i^{-1} \mathbf{V} \sum_r N_{ir} \Delta \boldsymbol{\lambda}_r \tag{51}$$

Define the spatial transport-response operator as

$$\mathbf{H}_i = \mathbf{A}_i^{-1} \mathbf{V} \tag{52}$$

Then

$$\Delta \mathbf{c}_i = \mathbf{H}_i \sum_r \nu_{ir} \Delta \boldsymbol{\lambda}_r \tag{53}$$

or, for cell P,

$$\Delta c_{i.P} = \sum_Q (\mathbf{H}_i)_{PQ} \sum_r \nu_{ir} \Delta \lambda_{r.Q} \tag{54}$$

Thus, an equilibrium-source correction in cell Q can affect the concentration in cell P through spatial transport. This coupling is retained when calculating the equilibrium-source Newton correction. The response operator is applied by solving the frozen transport system; neither the inverse transport matrix nor the full response matrix is

explicitly assembled. Fixed boundary forcing has zero perturbation, while the boundary contributions to the frozen operator and coupled-patch communication are retained.
For preconditioning, a local approximation is obtained by neglecting the off-diagonal entries of the transport matrix. The perturbation equation then reduces to

$$a_{i.P}\Delta c_{i.P} \approx V_P \sum_r \nu_{ir}\Delta\lambda_{r.P} \tag{55}$$

where $a_{i.P}$ is the stored transport diagonal coefficient. Defining

$$S_{i.P} = \frac{V_P}{a_{P.i}} \tag{56}$$

gives the local response approximation

$$\Delta c_{i.P} \approx S_{i.P} \sum_r \nu_{ir}\Delta\lambda_{r.P} \tag{57}$$

Collecting the species responses within cell P,

$$\mathbf{S}_p = \mathrm{diag}(S_{1.P}, ..., S_{N.P}) \tag{58}$$

the local approximation becomes

$$\Delta\mathbf{c}_P \approx \mathbf{S}_P\mathbf{N}\Delta\boldsymbol{\lambda}_P \tag{59}$$

where $\mathbf{N}$ is the species-by-reaction stoichiometric matrix. This local approximation is used to construct the Newton preconditioner; the Newton operator itself uses the spatially coupled transport response $\mathbf{H}_i$.

A.1.2 General solution form

For a system with N_s participating species and N_r equilibrium reactions, the concentrations within cell P are collected in

$$\mathbf{c}_P = \begin{bmatrix} c_{1,P} \\ c_{2,P} \\ \vdots \\ c_{N_s,P} \end{bmatrix} \tag{60}$$

The stoichiometric coefficients are collected in the species-by-reaction matrix

$$\mathbf{N} = \begin{bmatrix} \nu_{11} & \nu_{12} & \cdots & \nu_{1N_r} \\ \nu_{21} & \nu_{22} & \cdots & \nu_{2N_r r} \\ \vdots & \vdots & \ddots & \vdots \\ \nu_{N_s 1} & \nu_{N_s 2} & \cdots & \nu_{N_s N_r} \end{bmatrix} \tag{61}$$

where positive coefficients denote products and negative coefficients denote reactants. Using the equilibrium equation

$$K = \frac{\prod_{\nu_i>0} c_i^{\nu_i}}{\prod_{\nu_i<0} c_i^{-\nu_i}} \tag{62}$$

the equilibrium condition can be expressed as a residual F_r. Within cell P, these residuals form the vector

$$\mathbf{F}_P(\mathbf{c}_P) = \begin{bmatrix} F_1(\mathbf{c}_P) \\ F_2(\mathbf{c}_P) \\ \vdots \\ F_{N_r}(\mathbf{c}_P) \end{bmatrix} = \begin{bmatrix} \prod_{\nu_{i1}>0} c_{i.P}^{\nu_{i1}} - K_1 \prod_{\nu_{i1}<0} c_{i.P}^{-\nu_{i1}} \\ \prod_{\nu_{i2}>0} c_{i.P}^{\nu_{i2}} - K_2 \prod_{\nu_{i2}<0} c_{i.P}^{-\nu_{i2}} \\ \vdots \\ \prod_{\nu_{iN_r}>0} c_{i.P}^{\nu_{iN_r}} - K_{N_r} \prod_{\nu_{iN_s}<0} c_{i.P}^{-\nu_{iN_r}} \end{bmatrix} = \mathbf{0} \tag{63}$$

The corresponding concentration Jacobian is

$$\mathbf{J}_{c.P} = \frac{\partial\mathbf{F}_P}{\partial\mathbf{c}_P} \tag{64}$$

A first-order Taylor expansion gives

$$\mathbf{F}_P(\mathbf{c}_P + \Delta\mathbf{c}_P) \approx \mathbf{F}_P(\mathbf{c}_P) + \mathbf{J}_{c.P}\Delta\mathbf{c}_P \tag{65}$$

Requiring the linearized residual to vanish yields

$$\mathbf{F}_P + \mathbf{J}_{c.P}\Delta\mathbf{c}_P \approx 0 \tag{66}$$

The concentration response is substituted into the expression

$$\mathbf{J}_{c.P}\mathbf{H}_i \sum_r \nu_{ir}\Delta\boldsymbol{\lambda}_r = -\mathbf{F}_P \tag{67}$$

From this the equilibrium-source Jacobian is defined

$$\mathbf{J}_\lambda = \mathbf{J}_c\mathbf{H}_i \sum_r \nu_{ir} \tag{68}$$

Collecting the source corrections and residuals over all reactions and cells gives the global Newton system

$$\mathbf{J}_\lambda\Delta\boldsymbol{\lambda} = -\mathbf{F} \tag{69}$$

This system contains $N_r N_c$ unknown source corrections that are solved to obtain the equilibrium source field update

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha_\lambda\Delta\boldsymbol{\lambda} \tag{70}$$

Here, $\boldsymbol{\lambda}$ represents the unknown volumetric equilibrium reaction rates, and $\Delta\boldsymbol{\lambda}$ is their Newton correction. α_λ is relaxation factor. The relaxation factor is dynamically found using the fraction to the boundary limiter

$$\alpha_\lambda = \min\left[1.0,\, \min_{i.P}\left(\eta\frac{c_{i.P}}{-\Delta c_{i.P}}\right)\right], \quad 0 \le \eta \le 1 \tag{71}$$

The parameter η specifies the permitted fraction of the distance to the concentration boundary $c_i = 0$. The minimum is taken over all species and cells throughout the domain.

A.1.3 Stoichiometry derived residual and Jacobian

For a general setup with minimal setup on the user side it is shown how the equilibrium constraint equation and concentration Jacobian can be obtained from stoichiometry coefficients and equilibrium constants. Using the general equilibrium equation

$$K = \frac{\prod_{\nu_i>0} c_i^{\nu_i}}{\prod_{\nu_i<0} c_i^{-\nu_i}}$$

The following is defined

$$P_r = \prod_{\nu_i>0} c_i^{\nu_i}, \quad R_r = \prod_{\nu_i<0} c_i^{-\nu_i} \tag{72}$$

The equilibrium residual becomes

$$F_r = P_r - K_r R_r = 0 \tag{73}$$

For the Jacobian

$$\frac{\partial F_r}{\partial c_i} = \begin{cases} \nu_{ir} c_i^{\nu_{ir}-1} \prod_{\substack{j\neq i \\ \nu_{jr}>0}} c_j^{\nu_{jr}}, & \text{for} \quad \nu_{ir} > 0, \\ K_r \nu_{ir} c_i^{-\nu_{ir}-1} \prod_{\substack{j\neq i \\ \nu_{jr}<0}} c_j^{-\nu_{jr}}, & \text{for} \quad \nu_{ir} < 0, \\ 0, & \text{for} \quad \nu_{ir} = 0 \end{cases} \tag{74}$$

A.2 Electro-neutral potential formulation

Potential is found through current conservation with current equation

$$\mathbf{j} = F \sum_i z_i \mathbf{N}_i \tag{75}$$

Where "$\mathbf{N}_i$" is the species flux

$$\mathbf{N}_i = \mathbf{u}c_i - D_i \nabla c_i - \frac{z_i F}{RT} D_i c_i \nabla \phi \tag{76}$$

and charge density

$$\rho_e = F \sum_i z_i c_i \tag{77}$$

Including reaction the full current conservation equation becomes

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \mathbf{j} = F \sum_i z_i R_i \tag{78}$$

This expands to

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \left(F \mathbf{u} \sum_i z_i c_i \right) - \nabla \cdot \left(F \sum_i z_i D_i \nabla c_i \right) - \nabla \cdot \left(\frac{F^2}{RT} \sum_i z_i^2 D_i c_i \nabla \phi \right) = F \sum_i z_i R_i \tag{79}$$

From this potential can be solved such that current is conserved, however a solution that additionally enforce electro-neutrality is desired. For this an additional constraint of no charge concentration is imposed

$$\rho_e = 0 \Rightarrow \sum_i z_i c_i = 0 \tag{80}$$

Additionally as reactions are charge conserving the following can be imposed too

$$\sum z_i R_i = 0 \tag{81}$$

With this the charge density, advection and reaction terms become zero and the current conservation simplifies to a divergence free current equation

$$\nabla \cdot \mathbf{j} = 0 \Rightarrow \nabla \cdot \left(-F \sum z_i D_i \nabla c_i - \frac{F^2}{RT} \sum z_i^2 D_i c_i \nabla \phi \right) = 0 \tag{82}$$

From this diffusive flux is defined as

$$\mathbf{J}_D = -F \sum z_i D_i \nabla c_i \tag{83}$$

and conductivity

$$\kappa = \frac{F^2}{RT} \sum z_i^2 D_i c_i \tag{84}$$

Substituting these into the current conservation equation gives

$$\nabla \cdot (\mathbf{J}_D - \kappa \nabla \phi) = 0 \tag{85}$$

or rearranged to elliptic form

$$\nabla \cdot (\kappa \nabla \phi) = \nabla \cdot \mathbf{J}_D \tag{86}$$

The equation is discretized over the computational domain using the finite-volume method. For a given transported concentration state, $\mathbf{J}_D$ and κ are known and the potential equation is linear in ϕ. The discretized equation can therefore be written generally as

$$\mathbf{A}_\phi \mathbf{\Phi} = \mathbf{b}_D \tag{87}$$

In the Picard loop, the transported concentration state at iteration k is used to assemble the conductivity operator $\mathbf{A}_\phi^{(k)}$ and source vector $\mathbf{b}_D^{(k)}$. The potential that satisfies current conservation for this concentration state is denoted $\mathbf{\Phi}^*$ and satisfies

$$\mathbf{A}_\phi^{(k)} \mathbf{\Phi}^* = \mathbf{b}_D^{(k)} \tag{88}$$

The current potential $\mathbf{\Phi}^{(k)}$ was used to obtain the transported concentration state and does not generally satisfy this newly assembled equation. The current-conserving potential is therefore written as

$$\mathbf{\Phi}^{*} = \mathbf{\Phi}^{(k)} + \Delta\mathbf{\Phi} \tag{89}$$

The goal is to find the correction to the current potential $\Delta\mathbf{\Phi}$ that satisfies current conservation. Substituting Φ^{*} for the current potential gives

$$\mathbf{A}_{\phi}^{(k)} \left(\mathbf{\Phi}^{(k)} + \Delta\mathbf{\Phi} \right) = \mathbf{b}_{D}^{(k)} \tag{90}$$

This rearranges to

$$\mathbf{A}_{\phi}^{(k)} \Delta\Phi = \mathbf{b}_{D}^{(k)} - \mathbf{A}_{\phi}^{(k)} \Phi^{(k)} \tag{91}$$

The right-hand side can be defined as the current-conservation equation residual

$$\mathbf{r}_{\phi}^{(k)} = b_{D}^{(k)} - \mathbf{A}_{\phi}^{(k)} \Phi^{(k)} \tag{92}$$

Substituting in the residual gives

$$\mathbf{A}_{\phi}^{(k)} \Delta\Phi = \mathbf{r}_{\phi}^{(k)} \tag{93}$$

resulting in a linear matrix system that can be solved for the potential correction $\Delta\mathbf{\Phi}$ from which the potential can be updated

$$\mathbf{\Phi}^{(k+1)} = \mathbf{\Phi}^{(k)} + \alpha_{\phi} \Delta\mathbf{\Phi} \tag{94}$$

This correction potential is applied to the next Picard step and electro-neutrality is evaluated again. In this manner, species transport and potential correction are iterated until the concentration and potential fields are mutually consistent with both electro-neutrality and current conservation.

A.3 Boundary conditions

This section relies on the following sign conventions

$$
\begin{aligned}
\mathbf{n}: &\quad \text{control-volume outward-pointing unit normal vector} \\
j_{app}: &\quad \text{magnitude of applied current density,} \quad j_{app} \geq 0 \\
\mathbf{N}_i \cdot \mathbf{n} > 0: &\quad \text{species flux out of control-volume} \\
\mathbf{j} \cdot \mathbf{n} > 0: &\quad \text{ionic current out of control-volume} \\
\\
\nu_i > 0: &\quad \text{production of aqueous species } i \\
\nu_i < 0: &\quad \text{consumption of aqueous species } i \\
\nu_e > 0: &\quad \text{production of electrons} \\
\nu_e < 0: &\quad \text{consumption of electrons}
\end{aligned}
$$

A.3.1 Electrode boundaries

For a species "i" with flux

$$\mathbf{N}_i = \mathbf{u}c_i - D_i \nabla c_i - \frac{z_i F}{RT} D_i c_i \nabla \phi \tag{95}$$

At an impermeable solid electrode $\mathbf{u} \cdot \mathbf{n} = 0$, the surface normal flux is

$$\mathbf{N}_i \cdot \mathbf{n} = -D_i \nabla c_i \cdot \mathbf{n} - \frac{z_i F}{RT} D_i c_i \nabla \phi \cdot \mathbf{n} \tag{96}$$

OpenFOAM sign conventions follows that $\mathbf{n} > 0$ describes flux out of the control volume and if a positive stoichiometric coefficient $\nu_i > 0$ means production of species i then the faradaic production at an electrode must follow

$$N_{i.e} = -\frac{\nu_i j_{app}}{|\nu_e| F} \tag{97}$$

where j_{app} is the magnitude of the galvanostaticly applied current, and ν_e is the stoichiometric coefficient for electrons participating in the reaction. Hence, if $\nu_i > 0$ then $N_{i.e} < 0$ and species is produced into the control-volume and oppositely when $\nu_i < 0$ species is consumed. This faradic flux, equations 97, must be consistent with the Nernst-Planck flux, equation 96

$$\mathbf{N}_i \cdot \mathbf{n} = N_{i.e} \tag{98}$$

A.3.2 Concentration gradient

To get the concentration gradient at the boundary equations 96 and 97 are inserted into equation 98

$$-D_i \nabla c_i \cdot \mathbf{n} - \frac{z_i F}{RT} D_i c_i \nabla \phi \cdot \mathbf{n} = -\frac{\nu_i j_{app}}{|\nu_e| F} \tag{99}$$

In this $\nabla c_i \cdot \mathbf{n}$ is isolated to form a robin boundary condition

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{app}}{D_i |\nu_e| F} - \frac{z_i F}{RT} c_i \nabla \phi \cdot \mathbf{n} \tag{100}$$

A.3.3 Potential gradient

The potential gradient is derived from the current

$$\mathbf{j} = F \sum_i z_i \mathbf{N}_i \tag{101}$$

on electrode normal component form

$$\mathbf{j} \cdot \mathbf{n} = F \sum_i z_i (\mathbf{N}_i \cdot \mathbf{n}) \tag{102}$$

From equation 98 it is given that $\mathbf{N}_i \cdot \mathbf{n} = N_{i.e}$ which is inserted

$$\mathbf{j} \cdot \mathbf{n} = F \sum_i z_i N_{i.e} \tag{103}$$

which expands to

$$\mathbf{j} \cdot \mathbf{n} = -\frac{j_{app}}{|\nu_e|} \sum_i z_i \nu_i \tag{104}$$

From the charge balance of the electrode half-reaction, given the electrode charge is fixed as $z_e = -1$ then

$$\sum_i z_i \nu_i + z_e \nu_e = 0 \Rightarrow \sum_i z_i \nu_i = \nu_e \tag{105}$$

This is substituted back into equation 104

$$\mathbf{j} \cdot \mathbf{n} = -\frac{\nu_e}{|\nu_e|} j_{app} \tag{106}$$

This shows the direction of the electrode normal current follows from the electron stoichiometry of the half-reaction. or an electron-consuming reaction, ($\nu_e < 0$), the normal current is positive and conventional ionic current leaves the electrolyte through the electrode boundary. For an electron-producing reaction, ($\nu_e > 0$), the normal current is negative and conventional ionic current enters the electrolyte through the electrode boundary. For simplicity an electrode aware current density is defined as

$$j_{n.app} \equiv \mathbf{j} \cdot \mathbf{n} = -\frac{\nu_e}{|\nu_e|} j_{app} \tag{107}$$

Expanding equation 101 with 96 gives the current equation

$$\mathbf{j} = \mathbf{J}_D - \kappa \nabla \phi \tag{108}$$

with diffusive flux

$$\mathbf{J}_D = -F \sum_i z_i D_i \nabla c_i \tag{109}$$

and conductivity

$$\kappa = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i \tag{110}$$

This can be rewritten as

$$(\mathbf{J}_D - \kappa \nabla \phi) \cdot \mathbf{n} = j_{n.app} \tag{111}$$

where the potential gradient can be isolated to a Neumann boundary condition

$$\nabla \phi \cdot \mathbf{n} = \frac{\mathbf{J}_D \cdot \mathbf{n} - j_{n.app}}{\kappa} \tag{112}$$

A.3.4 Picard coupling

The concentration and potential boundary conditions are physically coupled. The concentration gradient depends on the electric potential gradient through migration, while the potential gradient depends on the concentration fields through the diffusive current and electrolyte conductivity. These conditions describe the state that must be satisfied by the converged solution; they are not solved as a separate simultaneous nonlinear boundary system.
Consistent with the segregated Picard treatment used in the domain equations, the electrode coupling is lagged between concentration transport and potential correction. During Picard iteration k, the species transport equations are solved using the currently available potential field $\phi^{(k)}$, including its existing electrode-normal gradient. The electrode concentration condition is therefore evaluated as

$$\nabla c_i^{(k+1)} \cdot \mathbf{n} = \frac{\nu_i j_{app}}{D_i |\nu_e| F} - \frac{z_i F}{RT} c_i^{(k+1)} \nabla \phi^{(k)} \cdot \mathbf{n} \tag{113}$$

The potential gradient appearing in the migration contribution is consequently frozen during the concentration transport step. The concentration boundary condition does not request or calculate a new potential solution. Introducing

$$A_I = \frac{\nu_i j_{app}}{|\nu_e| F D_i}, \quad \beta_i^{(k)} = \frac{z_i F}{RT} (\nabla \phi^{(k)} \cdot \mathbf{n}) \tag{114}$$

gives the Robin condition

$$\nabla c_i^{(k+1)} \cdot \mathbf{n} = A_i - \beta_i^{(k)} c_{b.i}^{(k+1)} \tag{115}$$

With the boundary-normal finite-volume gradient

$$\nabla c_i^{(k+1)} = \delta(c_{b.i}^{(k+1)} - c_{p.i}^{(k+1)}) \tag{116}$$

where $\delta = 1/\Delta n$, the boundary concentration becomes

$$c_{b.i}^{(k+1)} = \frac{\delta c_{p.i}^{(k+1)} + A_i}{\delta + \beta_i^{(k)}} \tag{117}$$

Thus the concentration boundary is implicit with respect to its own boundary concentration, but explicitly lagged with respect to the potential gradient. This is the boundary analogue of the Picard treatment used for the internal transport equations. After the concentration transport and equilibrium correction have produced the updated concentration state $c_i^{(k+1)}$, the potential is corrected from current conservation. The diffusive flux and electrolyte conductivities are updated using the updated concentrations

$$\mathbf{J}_D^{(k+1)} = -F \sum_i z_i D_i \nabla c_i^{(k+1)} \tag{118}$$

$$\kappa^{(k+1)} = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{(k+1)} \tag{119}$$

Resulting in the potential boundary gradient

$$\nabla \phi^{(k+1)} \cdot \mathbf{n} = \frac{\mathbf{J}_D^{(k+1)} \cdot \mathbf{n} - j_{n.app}}{\kappa^{(k+1)}} \tag{120}$$

The newly corrected potential gradient is not fed recursively back into the concentration boundary condition during the same transport solve. Instead, it becomes the frozen potential-gradient state used by the concentration boundary condition in the next Picard iteration. The electrode coupling therefore follows the sequence

$$\phi^{(k)} \to c_i^{(k+1)} \to \mathbf{J}_D^{(k+1)},\ \kappa^{(k+1)} \to \phi^{(k+1)}$$

with physical time held fixed. At Picard convergence the lag disappears and the concentration and potential boundary conditions simultaneously satisfy the original electrode flux and current-conservation relations.

A.4 MES setup

A.4.1 Reaction simplifications

Full reaction scheme

$$\begin{aligned} r_W: &\quad H_2O \rightleftharpoons H^+ + OH^- \\ r_1: &\quad \mathrm{CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+} \\ r_2: &\quad HCO_3^- \rightleftharpoons CO_3^{2-} + H^+ \\ r_3: &\quad CO_2 + OH^- \rightleftharpoons HCO_3^- \\ r_4: &\quad CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^- \end{aligned}$$

The CO_2-bicarbonate equilibrium r_1 consists of a lumped hydration step $r_{1.h}$ and acid dissociation step $r_{1.a}$

$$r_1: \mathrm{CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+} \begin{cases} r_{1.h}: & \mathrm{CO_2 + H_2O \rightleftharpoons H_2CO_3} \\ r_{1.a}: & \mathrm{H_2CO_3 \rightleftharpoons HCO_3^- + H^+} \end{cases}$$

The CO_2 hydration step as well as the hydroxide path r_3 are both slow reactions and cannot comfortably be treated as instantaneous equilibrium projections and must be treated kinetically

$$\dot{r}_{1.h} = k_{f.1} c_{CO_2} - k_{b.1} c_{H_2CO_3} \tag{121}$$

$$\dot{r}_3 = k_{f.3} c_{CO_2} c_{OH^-} - k_{b.3} c_{HCO_3^-} \tag{122}$$

reactions $r_{1.h}$ and $r_{1.a}$ can be thermodynamically lumped into a single kinetic equation.

$$K_{1.2} = \frac{c_{HCO_3^-} c_{H^+}}{c_{H_2CO_3}} \Rightarrow c_{H_2CO_3} = \frac{c_{HCO_3^-} c_{H^+}}{K_{1.2}} \tag{123}$$

$$\dot{r}_1 = k_{f.1} c_{CO_2} - \frac{k_{b.1}}{K_{1.2}} c_{HCO_3^-} c_{H^+} \tag{124}$$

Effectively eliminating H_2CO_3 as a variable and making CO_2 treated entirely kinetically. Then the renaming equilibrium reactions are

$$K_W = c_{H^+} c_{OH^-} \tag{125}$$

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \tag{126}$$

$$K_4 = \frac{c_{HCO_3^-} c_{OH^-}}{c_{CO_3^{2-}}} \tag{127}$$

Where r_4 can be shown to be redundant. The equilibrium equations for water and bicarbonate-carbonate are rewritten

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \Rightarrow \frac{c_{HCO_3^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+}}{K_2} \tag{128}$$

$$K_W = c_{H^+} c_{OH^-} \Rightarrow c_{OH^-} = \frac{K_W}{c_{H^+}} \tag{129}$$

The rewritten forms are multiplied and H^+ cancel out on the right-hand side

$$\frac{c_{HCO_3^-} c_{OH^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+} K_W}{K_2 c_{H^+}} = \frac{K_W}{K_2} \tag{130}$$

Showing the left-hand side is exactly the equilibrium constant for r_4

$$K_4 = \frac{K_W}{K_2} \tag{131}$$

making it a linear combination of the other equilibrium reactions and thereby redundant as they are not resolved kinetically but projected.

A.4.2 Full reaction scheme

With the simplifications given above the reaction scheme can be split into slow and fast reactions. The slow reactions being solved using the kinetic approach

$$\dot{r}_1 = k_{f.1} c_{CO_2} - \frac{k_{b.1}}{K_{1.2}} c_{HCO_3^-} c_{H^+} = k_{f.1} c_{CO_2} - k^*_{b.1} c_{HCO_3^-} c_{H^+}$$

$$\dot{r}_3 = k_{f.3} c_{CO_2} c_{OH^-} - k_{b.3} c_{HCO_3^-}$$

using $k^*_{b.1} = \frac{k_{b.1}}{K_{1.2}}$. And fast equilibrium reactions

$$K_W = c_{H^+} c_{OH^-}$$

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}}$$

A.4.3 Equilibrium Newton system

From the fast equilibrium reactions

$$H_2O \rightleftharpoons H^+ + OH^- \tag{132}$$

$$HCO_3^- \rightleftharpoons CO_3^{2-} + H^+ \tag{133}$$

the concentration vector is defined

$$\mathbf{c} = \begin{bmatrix} c_{H^+} \\ c_{OH^-} \\ c_{HCO_3^-} \\ c_{CO_3^{2-}} \end{bmatrix} \tag{134}$$

and using reaction ordering (r_W, r_2), then the stoichiometric matrix becomes

$$\mathbf{N} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix} \tag{135}$$

and constraint equations

$$\mathbf{F}(\mathbf{c}) = \begin{bmatrix} c_{H^+} c_{OH^-} - K_W \\ c_{CO_3^{2-}} c_{H^+} - K_2 c_{HCO_3^-} \end{bmatrix} \tag{136}$$

then the concentration Jacobian is defined

$$\mathbf{J}_c = \frac{\partial \mathbf{F}}{\partial \mathbf{c}} = \begin{bmatrix} c_{OH^-} & c_{H^+} & 0 & 0 \\ c_{CO_3^{2-}} & 0 & -K_2 & c_{H^+} \end{bmatrix} \tag{137}$$